Consolidated Drones Capabilities Report

Country	Model	Length	Wingspan	Range	Speed	Endurance	Payload
Europe	BAE Systems Taranis	N/A	N/A	N/A	1126+ km/h	N/A	N/A
The United States	RQ-170 Sentinel	N/A	N/A	N/A	949.3 km/h	N/A	N/A
The United States	Boeing MQ-25 Stingray	N/A	N/A	N/A	933.4 km/h	N/A	N/A
The United States	R-Q 4 Global Hawk	14.5 m	39.8 m	22,800 km	357 km/h	34/h	1, 360 kg
The United States	MQ-9 Reaper	11 m	20.1 m	1, 850 km	444.48 km/h	N/A	1, 701 kg
The United States	Predator C (MQ-20) Avenger	13 m	20 m	5, 835 km	740 km/h	20/h	2, 948 kg
The United States	MQ-5B Hunter	7 m	8.84 m	N/A	222 km/h	30/h	N/A
The United States	Northrop Grumman X-47B	11.62 m	18.9 m	3890 km	N/A	12/h	2000 kg
The United States	XQ-58A	9 m	8.2 m	5556 km	882 km/h	N/A	272 kg (internal)/ 272kg (external)
The United States	Switchblade 600	130 cm	N/A	40 km	185 km/h	40/min	15 kg
The United States	MQ-1C-Gray Eagle	9 m	17 m	N/A	310 km/h	25/h	227-262 kg
Turkey	Bayraktar TB-2	6.5 m	12 m	300 km	222 km/h	27/h	150 kg

Country	Model	Length	Wingspan	Range	Speed	Endurance	Payload
Turkey	Bayraktar TB-3	8.35 m	14 m	N/A	300 km	24/h	280 kg
Turkey	Bayraktar-Akinci	12.2 m	20 m	7,500 km	360 km/h	24/h	1,500 kg
Turkey	Aksungur	11.6 m	24 m	N/A	N/A	50/h	N/A
Turkey	Bayraktar DIHA (VTOL)	1.5 m	5 m	150 km	150/h	12/h	5 kg
Turkey	Kizilelma	14.7 m	10 m	930 km	900 km/h	5/h	1,500 kg
China	Wing Loong-1	9 m	14 m	N/A	280 km/h	20/h	200 kg
China	Wing Loong-2	11 m	20.5 m	4,000 km	370 km/h	20/h	480 kg
China	Wing Loong-3	12.2 m	24 m	10,000 km	402 km/h	40/h	2, 300 kg
China	CASC CH-5 Rainbow	11.2 m	26 m	10,000 km	480 km/h	60/h	1,200 kg
China	Gonji -11 (GJ-11)	11.65 m	14 m	4,000 km	1000 km/h	N/A	N/A
Israel	Heron TP	14 m	26 m	1000 km	220 km/h	30/h	2,700 kg
Israel	Point Blank	0.91 m	0.8 m	10 km	186 km/h	20/min	2 kg
Israel	Heron	8.5 m	16.6 m	1000 km	111-148 km/h	45/h	470 kg
Israel	Hermes 450	6.1 m	10.5 m	200 km	176 km/h	17-30/h	180 kg
Israel	Hermes 900 (aka Kochav)	8.3 m	15 m	9,000 km	220 km/h	36/h	350 kg
Iran	Ababil 3	4.5 m	6.5 m	150 km	300 km/h	8/h	1.7 kg
Iran	Arash (aka Kian 2)	4.5 m	4 m	2000 km	480 km/h	30/h	30 kg

Country	Model	Length	Wingspan	Range	Speed	Endurance	Payload
Iran	Fetros	8.25 m	16 m	1700 – 2000 km	250 km/h	30/h	N/A
Iran	Kaman 22	6.5 m	17 m	3000 km	N/A	24/h	300 kg
Iran	Mohajer 6	5.67 m	10 m	2000 km	200 km/h	12/h	40 kg
Iran	Shahed 129	8 m	16 m	1700 km	200 km/h	24/h	400 kg
Iran	Shahed 136	3.5 m	2.5 m	2000 km	185 km/h	N/A	50 kg
Iran	Shahed 191 (aka Saegheh 2)	2.7 m	7.31 m	1500 km	3500 km/h	5 /h	50 kg
Iran	Yasir (aka Qods Yasir/Sayed 2)	1.19 m	3 m	450 km	120 km/h	8-20/h	N/A
Iran	Mohajer 10	Unknown	Unknown	2000 km	210 km/h	24/h	300 kg
Europe	Dassault nEUROn	9.5 m	12.5 m	N/A	980 km/h	N/A	N/A
Russia	Sukhoi SU-70	11.65 m	20 m	6,000 km	1,400 km/h	N/A	2,000 kg
Germany and Spain	EADS Barracuda	8.25 m	7.22 m	200 km	1040 km/h	17/h	300 kg
Australia	MQ-28 Ghost Bat	11.7 m	7.3 m	3700 km	296 km/h	16/h	N/A
France	AAROK	14 m	22 m	N/A	463 km/h	24-plus/h	2,720 kg
Taiwan	Albatross II	6.5 m	14.97 m	2255 km	205 km/h	20 hours	250 kg

Propulsion Systems: Jet-powered engines drastically increase the price compared to off-the-shelf commercial piston engines.
Advanced Sensors and Avionics: High-resolution electro-optical/infrared (EO/IR) cameras, synthetic aperture radar (SAR), and LiDAR systems are incredibly expensive to develop, miniaturize, and integrate.

Secure Communications: Anti-jamming antennas (like CRPA arrays), satellite uplinks for beyond-visual-line-of-sight (BVLOS) operations, and highly encrypted datalinks require sophisticated engineering.
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Materials: Radar-absorbent coatings, carbon-fiber composites for reduced weight, and specialized chassis shapes designed to minimize radar cross-sections add significant manufacturing costs.
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AI and Autonomy: Systems that allow a drone to independently identify targets using optical recognition, adjust flight paths dynamically, or operate in synchronized "swarms" require heavy investment in software and processing hardware
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Military Drones

The high acquisition and operational costs of Unmanned Aerial Vehicles (UAVs) are primarily driven by the sophisticated integration of propulsion technology, high-grade hardware components, and complex flight control software required for safe and reliable mission execution.

The developmental lifecycle necessitates a multi-disciplinary engineering framework comparable to major aerospace corporations, encompassing specialists in electrical systems, mechanical structures, fluid dynamics, autonomous software architecture, and industrial design.

Research and Development (R&D) expenditures for industry-leading manufacturers remain substantial. Furthermore, global supply chain volatility and the increasing scarcity of advanced semiconductors and System-on-Chip (SoC) architectures have significantly inflated unit manufacturing costs in recent fiscal periods.

The democratization of drone warfare

When Ukrainian soldiers began destroying Russian tanks with inexpensive quadcopters, the economics of warfare shifted suddenly. At the same time, Iranian drones appeared on Middle Eastern battlefields ranging from Yemen to Lebanon, while non-state actors such as the Houthis and Hezbollah started their own aerial attacks.

Airpower, formerly the sole domain of advanced militaries, is fast becoming available to a considerably broader spectrum of entities. The proliferation of drones, which are increasingly powered by artificial intelligence, is altering combat in ways that governments are only beginning to comprehend.

For most of the twentieth century, airpower was one of the most costly components of military strength. Fighter jets, strategic bombers, and advanced missile systems necessitated massive financial investments, highly skilled personnel, and sophisticated logistical support networks. Only great military forces had the ability to sustain these capabilities.

Drone technology is changing the equation.

Commercial drones are readily available and becoming increasingly powerful. With little changes, they can be used for monitoring, targeting, and even direct attacks. Artificial intelligence is also helping to improve these systems. Drones may use AI to recognise objects, track targets, avoid obstacles, and navigate complex settings. Analysts often refer to the marriage of AI and drones as one of the most significant technological developments in modern warfare, particularly in debates about autonomous weapons systems and emerging military technologies.

The end result is a new type of aerial capability that is less expensive, more adaptable, and much more accessible than traditional airpower.

This change is already evident in some wars.

In Yemen, the Houthi movement has extensively deployed drones to assault military targets and crucial infrastructure in Saudi Arabia and the Gulf. Drone strikes on oil installations and airports highlighted how low-cost unmanned technologies might impose strategic consequences on much wealthier countries. Analysts have documented how the Houthis have constructed a developing drone arsenal capable of long-range assaults using very simple technology, as revealed in a West Point Combating Terrorism Center on the Houthi drone program.

Hezbollah has also improved its drone capabilities with Iranian assistance. Drones have been deployed for surveillance activities around Israel's borders, and unmanned technologies are rapidly being integrated into military plans. Israeli authorities have repeatedly warned that Hezbollah's drone weapons may be a significant factor in any future conflict between the two sides. Reports on Hezbollah's developing drone capacity and Iranian help have received a lot of attention in evaluations of Iran’s expanding unmanned aerial vehicle strategy.

Militant terrorist organisations have also shown how quickly drone technology spreads. During its territorial control over Iraq and Syria, ISIS used commercially accessible drones to conduct reconnaissance and drop explosives on hostile forces. These devices were rudimentary in comparison to the military drones deployed by advanced powers, but they were effective enough to disrupt battlefield operations and highlighted how quickly civilian technology might be weaponized. Researchers looking into the Islamic State's drone program discovered how ISIS modified off-the-shelf drones for war usage.

Perhaps the most striking example of drone warfare today is the war in Ukraine.

Both Ukrainian and Russian military rely heavily on drones for reconnaissance, artillery targeting, and direct attacks. Ukrainian soldiers and volunteer units have integrated small commercial drones into nearly every aspect of battlefield operations. Cheap quadcopters are used to pinpoint enemy positions, direct artillery fire, and carry explosives to armoured vehicles. According to analysts at the Royal United Services Institute (RUSI), drones have become one of the conflict's distinguishing aspects.

Artificial intelligence is rapidly being used to evaluate drone footage, identify targets, and enhance battlefield coordination. In several occasions, drones costing only a few thousand dollars damaged tanks and military equipment worth millions of dollars. The enormous disparity between cost and efficacy is changing the economics of combat.

Another revealing example is emerging along the Afghanistan–Pakistan border.

Pakistan has a nuclear arsenal, a huge conventional military, and is investing heavily in modern technologies such as artificial intelligence and surveillance. Yet militant groups operating in the Afghanistan–Pakistan region have increasingly experimented with drones for reconnaissance and potential attacks. Recent reporting has highlighted rising tensions and cross-border clashes between Pakistan and Afghanistan, demonstrating how the spread of low-cost drone technology could change the dynamics of conflict between stronger states and weaker actors, as reported in coverage of Pakistan–Afghanistan border tensions and militant activity.

The ramifications of this transformation are substantial. Pakistan's military remains far more strong than the militant organisations operating in the region. However, with the widespread use of drones, even loosely organised insurgent networks can increasingly pose a threat to military and border infrastructure. This parallels a larger trend in modern warfare, in which technical spread enables weaker actors to attack bigger ones in novel ways.

Regional powers are also expanding their drone capabilities.

Iran has one of the largest drone programmes in the Middle East. Iranian drones, such as the Shahed series, have emerged in a number of regional conflicts and have been given to friendly forces throughout the area. These techniques enable Tehran to exert influence through allies and proxies rather of depending just on conventional military deployments. Analysts studying Iran's growing drone industry and regional strategy believe that these systems have become an important component of Tehran's military plan.
Drones have played an important part in surveillance operations and military strikes in recent confrontations between Iran, the United States, and Israel. Both Washington and Jerusalem have made significant investments in advanced unmanned systems designed to operate in disputed situations. Artificial intelligence is increasingly integrated into these platforms to improve targeting, navigation, and operational coordination.

ABSTRACT

The article examines the recent and rapid proliferation of cheap, unsophisticated "kamikaze" drones, as well as the legal and moral challenges they present in modern warfare. As countries increasingly rely on these cost-effective yet crude drones, the risks to civilian safety and the potential for violations of international humanitarian law are heightened. The article highlights the development of drone technology and its impact on warfare. It critically examines the legal implications of using such drones, focusing on how their shortcomings can lead to violations of international humanitarian law. By analysing the incident involving Iranian drones in the April 2024 attack on Israel, the article underscores the failures of existing international regulatory frameworks. It calls for urgent international co-operation to establish comprehensive regulations that address challenges posed by new technologies and emphasises the need for effective monitoring and enforcement of international humanitarian law as warfare technology continues to advance rapidly.

KEYWORDS:

Low-cost drones
civilian safety and protection
remote warfare
kamikaze drones
international peace and security

international law

1. Introduction

Drone warfare has become a critical aspect of modern combat, with their use spreading rapidly.

Footnote

[1] This trend is expected to accelerate due to the drones' relative ease of availability and their low cost compared to other types of armaments.

Footnote

[2] Whilst the widespread use of drones is due in part to regulatory gaps, it is also driven by geopolitical factors, economic motivations, and rapid technological developments that have outpaced the international legal framework. Amongst the many types of drones, this paper specifically discusses kamikaze drones, a form of crewed aerial vehicle (UAV) designed for one-time use through direct impact on a target. While drones generally have diverse applications, kamikaze drones are distinct in that they are inexpensive, simple to deploy, and increasingly utilised in asymmetric conflicts. The proliferation of these drones – rather than high-quality UAVs – poses a distinct challenge under international law. Their immense potential for indiscriminate effects, particularly in urban areas, makes them an exceptional threat to civilian lives, warranting an investigation into their legal and humanitarian implications. The international community's divided response to drone regulation, driven by conflicting national security agendas and the reluctance of great powers to impose restrictions on their own capabilities, has hindered progress towards comprehensive regulation. This has led to the widespread use – and abuse – of drones in numerous conflict areas.

Footnote

[3] The passive approach towards both the development and deployment of drones has exacerbated the challenges in controlling these systems, ensuring that the international legal framework remains inadequate to regulate drones and other autonomous weapons systems in accordance with international law.

Recent conflicts, such as the Russia-Ukraine war and the Iranian drone attack on Israel on 13–14 April, 2024, highlight an alarming trend: nations like Iran, Russia, and the Democratic People's Republic of Korea (North Korea) continue to develop and deploy drones.

Footnote

[4] This is the case despite stringent sanctions imposed on these countries by the UN and others. In recent years, they have significantly enhanced their drone capabilities and exported these systems globally, posing significant threats to both national and international security.

Footnote

[5]

This trend highlights the broader challenges posed by the rapid advancement and widespread proliferation of drone technology. The growing use of low-cost, unsophisticated drones presents unique risks, particularly due to their weak guidance systems, which increase the likelihood of violations of international human law (IHL). The rapid advances in drone technology, combined with the potential for its misuse, heighten concerns about IHL compliance, as these systems are increasingly likely to be used in the commission of various IHL violations. Additionally, the proliferation of low-cost drones introduces challenges that go beyond regulatory concerns.

Footnote

[6] While states have the sovereign right to develop weapons, they must still adhere to established quality standards.

Footnote

[7] Unfortunately, the current international legal framework is insufficient, and decisive global action to address this issue is lacking. The reluctance of key states to engage in stricter regulation stems from their own reliance on drone technology for military operations, counter-terrorism efforts, and strategic deterrence.

Footnote

[8] Additionally, the economic benefits of drone exports make some states resistant to stronger international oversight and control. Without global consensus, attempts at regulation remain fragmented and ineffective, leading to the widespread production and distribution of low-quality drones that often struggle to accurately engage even legitimate targets due to their weak guidance systems.

Footnote

[9]

This issue is further complicated by the widespread accessibility of drone technology, allowing not only state actors, but also non-state groups, including proxies and terrorist organisations, to easily obtain these systems.

Footnote

[10] States can supply their proxies with low-quality, domestically produced drones, as demonstrated by Iran providing drones to Hezbollah.

Footnote

[11] The lack of effective international regulation and proactive measures exacerbates the risks associated with drone usage, raising significant concerns about global security, accountability, and the dangers posed to civilians.

In response to these growing concerns, this article identifies the need for international co-operation in establishing better regulatory mechanisms and more stringent quality standards to address these issues. Further regulation is necessary to mitigate the risks associated with drone proliferation, encourage compliance with IHL, and better safeguard civilian lives. This article, therefore, examines these challenges to underscore the urgent need for international action on these matters. Without such a framework, greater and more widespread proliferation is likely, leading to an increase in violations of both IHL and international human rights law (IHRL). To address these challenges, it is imperative to implement robust regulatory measures, monitor countries involved in drone manufacturing, and establish stringent quality standards for production.

Without standards and regulations to ensure their implementation, many problems will persist. One such issue is the continued deployment of low-quality drones that cannot accurately identify and engage targets, leading to multiple violations of international law, including IHL and IHRL. These violations are already occurring on the battlefield, where limited resources are driving the use of cheap drones with indiscriminate effects. The fear of drones malfunctioning and causing unintended harm, also contributes to an environment of insecurity amongst civilians, who face the real threat of lethal drones landing on them or their property during conflict due to insufficient targeting capabilities. Therefore, a multifaceted approach that combines international collaboration, a stringent regulatory framework, and rigorous quality control standards is critical to safeguarding civilian lives and upholding international legal norms.

This article uses the recent Iranian-Israeli conflict as a case study to support the argument that there is a need to regulate the manufacture, supply, and use of drones. The events in question occurred on 1 April 2024, when Israel launched an airstrike on what appeared to be an Iranian consulate building in Damascus, resulting in sixteen casualties, including members of the Islamic Revolutionary Guard Corps, three senior commanders, and two Syrian civilians.

Footnote

[12] This incident, along with escalating regional tensions, led to a retaliatory strike from Iran on 13–14 April 2024.

Footnote

[13] Iran's response involved the deployment of approximately 170 drones, along with 120 ballistic missiles and 30 cruise missiles.

Footnote

[14] Notably, this marked the first time the Islamic Republic openly acknowledged a direct attack on Israel originating from its own territory. Whilst Israel successfully intercepted most of these drones using its Iron Dome, a multi-layered air defence system with allied support, the attack highlighted both the evolving role of drones in modern warfare and the limitations of Iranian drone capabilities.

By analysing this conflict, the article bridges various disciplines, including law and engineering, to explore the implications of drone technology in contemporary warfare. The study assesses the technical aspects of the weapons used in this conflict, with a focus on target identification and engagement precision to minimise civilian risk. Additionally, the article examines how states exploit current gaps in international oversight and regulation of drone manufacturing. The absence of specific international standards and effective monitoring contributes to violations of IHL. By addressing these regulatory and technical gaps, the article seeks to draw lessons from the use of drones in conflict situations and propose measures to mitigate further civilian harm.

2. The rise of kamikaze drones: technological progress and ethical implications

Drone technology has a long history, tracing its roots back to the early twentieth century, with significant innovations emerging during World War I. Although earlier aerial devices such as balloons and kites were used for surveillance in the nineteenth century, these were not UAVs in the modern sense.

Footnote

[15] The true precursors to today's sophisticated UAVs – powered, aerial vehicles that do not carry a human operator, use aerodynamic forces for lift, and can fly autonomously, or be remotely piloted – began to take shape with the development of basic flying bombs and target drones during World War I. These early innovations in drone technology foreshadowed both their military utility and potential for misuse. As these basic tools evolved into sophisticated systems, they introduced new risks and had significant implications for modern warfare. Regarding the global impact of drones, it has been observed that:

> As an epoch-defining concept – one that captures a pivotal moment in modern warfare – drone warfare highlights many of the novel personal, political, strategic, tactical, and technological aspects that remote-controlled and increasingly autonomous aerial systems have introduced into 21st-century conflict.
>
> Footnote

[16]

In this context, one of the key advances in drone technology has been the emergence of kamikaze drones, or loitering munitions, which have exposed significant ethical and legal deficiencies in current international frameworks.
Footnote
[17] Unlike sophisticated UAVs equipped with extensive sensors and communication systems, these drones vary enormously in their technological capabilities. Some high-end loitering munitions possess advanced sensors and targeting systems, allowing them to identify and engage targets with precision. However, a distinct subset of kamikaze drones – particularly low-tech, mass-produced variants such as the Shahed-136 – lack such refinement, leading to greater risks of indiscriminate harm. These kamikaze drones are designed to remain airborne for extended periods, identify and engage targets with high precision, and then crash into the target, sacrificing the drone itself. Despite their seemingly straightforward design, these drones have underscored critical gaps in ethical considerations and international law regarding their regulation.

For decades, drones have emerged as transformative agents in modern warfare, fundamentally altering conflict dynamics. Their profound impact is due to three key features. Firstly, their compact size enhances their ability to evade radar detection, making them less visible and harder to intercept.

Footnote

[18] Secondly, drones are relatively inexpensive and simple to manufacture, lowering the barriers for many nations to incorporate them into their military arsenals.

Footnote

[19] Finally, drones have increased the feasibility of remote warfare – an approach that uses technology to engage or influence targets from a distance without requiring direct physical presence.

Remote warfare itself is not a new concept. Historically, it began with archery and progressed through various artillery developments. Today, other weapon systems, such as precision-guided missiles, have refined the practice, enabling

targeted strikes with greater accuracy. However, while precision is a defining characteristic of many high-tech UAVs, not all kamikaze drones offer this advantage. The mass proliferation of cheap, poorly guided models has resulted in significant accuracy and reliability issues in combat. This evolution highlights the progression from earlier remote weaponry systems to the current standards in modern combat.

Despite the long history of remote warfare technologies, kamikaze drones have garnered significant global attention, surpassing traditional remote warfare technologies like missiles. This heightened focus is due to several unique characteristics of kamikaze drones. While both missiles and kamikaze drones serve similar roles in delivering precise, destructive payloads with advanced guidance systems, the challenges posed by kamikaze drones are distinct.

Footnote

[20] Their lower cost, simplicity, and broad accessibility make them more vulnerable to misuse, particularly by non-state actors. In contrast, missiles require significant infrastructure and expertise, limiting their availability and ensuring more controlled use. This distinction is critical: whereas some high-tech loitering munitions are equipped with sophisticated targeting systems that help mitigate collateral damage, lower-end kamikaze drones – such as those deployed in various recent conflicts – lack these safeguards, increasing the risk to civilians as well as to other targets that may not be lawfully attacked.

The increasing deployment of kamikaze drones by countries such as Iran and Russia underscores growing concerns about the inadequacies of current international regulations governing drone use. Drones like Iran's Shahed-136 have caused significant civilian harm in conflicts, illustrating how emerging technologies can exacerbate civilian casualties and suffering.

Footnote

[21]

Although all long-range weapons carry a risk of imprecision, drones are specifically designed to offer greater accuracy. However, when poorly manufactured, or deployed without adequate targeting protocols, they lose this

advantage, leading to indiscriminate effects. Less sophisticated models, with their rudimentary guidance systems, lack the precision needed to strike reliably specific targets, often resulting in unintended death and damage.

Footnote

[22] This raises the critical question of whether such weapons, with their potential for indiscriminate effects, should be permitted.

As nations such as Iran, Israel, the United States, Russia, Turkey, and China continue to develop various types of kamikaze drones tailored to their strategic objectives, it is important to recognise the diversity among these systems.

Footnote

[23] Although all these drones are classified as kamikaze, they differ significantly in sophistication, guidance systems, and operational effectiveness. Unlike other remote warfare technologies, drones offer real-time surveillance and precise target verification, yet their improper use undermines these advantages. Understanding and classifying these differences is crucial, as they lead to vastly different outcomes on the battlefield. These distinctions must be taken into consideration when monitoring and regulating their use. The key guidance systems employed include:

- **GPS (Global Positioning System):** Provides precise navigation to specified predefined co-ordinates.
- Footnote
- [24] The level of sophistication in GPS systems varies; sophisticated drones utilise highly accurate GPS technology, while more basic models rely on less precise, cost-effective versions.
- Footnote
- [25]
- **GNSS (Global Navigation Satellite System):** Enables precise positioning and navigation by utilising satellite signals to determine the drone's location and trajectory.
- Footnote

- [26] The sophistication of GNSS systems can vary significantly: high-end drones incorporate multi-frequency, multi-constellation GNSS receivers for enhanced accuracy and resilience in challenging environments, whereas more basic models may rely on single-frequency receivers with limited capabilities.
- Footnote
- [27]
- **INS (Inertial Navigation System):** Tracks movement and orientation using accelerometers and gyroscopes, independent of external signals.
- Footnote
- [28] The accuracy of INS systems varies; advanced models offer higher reliability, while simpler versions may experience drift over time.
- Footnote
- [29]
- **Radar Homing:** Locks onto radar emissions from targets, making it effective against radar systems and electronic emitters.
- Footnote
- [30] Variations in sensitivity, range, and signal processing complexity influence both the cost and operational effectiveness of these systems.
- Footnote
- [31]
- **Optical Sensors:** Utilise cameras and visual sensors for real-time imaging and precision-strikes.
- Footnote
- [32] The sophistication of optical sensors ranges from basic cameras to advanced multi-spectral imaging systems, which provide greater resolution and targeting capability.
- Footnote
- [33]
- **Infrared Sensors:** Detects heat signatures, making them useful for night operations or low-visibility conditions.
- Footnote

- [34] The level of sophistication in infrared sensors varies, with more advanced versions typically providing higher resolution and greater sensitivity.
- Footnote
- [35]

Differences in guidance systems, particularly regarding accuracy and effectiveness, create distinct strategic roles for kamikaze drones. These roles reflect the varied needs and resources of the nations that develop and use them. Drone performance is influenced by the reliance on specific guidance systems, with some models heavily dependent on well-developed GPS systems, whilst others integrate INS with optical or infrared sensors for enhanced targeting. For example, sophisticated drones like the Harop (Israel) and SkyStriker (Israel) offer greater precision due to their sophisticated systems. However, this increased capability comes with higher costs and added complexity, raising concerns about accountability and potential civilian harm. In contrast, simpler drones like the Shahed-136 (Iran), which rely on basic GPS and minimal optical sensors, are more prone to inaccuracies and, therefore, more likely to cause collateral damage due to their limited precision and flexibility.

The following chart illustrates the variety of guidance systems among leading global kamikaze drones, highlighting the different technological approaches and their broader implications for modern warfare and international security (Table 1):

Table 1. Drones' Specs

Table Footnote

[a].

Download CSVDisplay Table

This analysis of various kamikaze drones and their guiding-systems reveals a spectrum of sophistication and associated risks. As noted, these differences have significant implications for their operational effectiveness and legal ramifications

under IHL. The legal issues surrounding kamikaze drones, particularly under IHL, focus on their compliance with the principles of distinction, proportionality, and necessity.

Footnote

[36] The sophistication of their guidance systems raises concerns about their ability to accurately distinguish between military targets and civilian infrastructure, or civilians, which is a fundamental requirement under IHL.

Footnote

[37] The risk of disproportionate harm to civilians, or civilian objects due to imprecise targeting, or autonomous decision-making processes presents another significant legal challenge.

Footnote

[38] States and non-state actors using these technologies must ensure adherence to IHL obligations to minimise civilian casualties and avoid indiscriminate attacks, or they risk potential liability under international law.

This article classifies these drones into three categories – High Sophistication, Moderate Sophistication, and Basic Sophistication – based on their guidance technologies. This novel classification of drones, not explicitly established in the existing literature, is introduced in this article to highlight the differences in technological sophistication. It is necessary the better to understand the diverse challenges these drones create in modern warfare and their impact on civilian safety. This classification emphasises that whilst all categories pose significant risks, drones in the basic sophistication category require particular scrutiny and oversight due to their higher likelihood of causing harm. Their low cost and ease of acquisition make them more accessible and prone to misuse, thus exacerbating threats to civilian safety.

High Sophistication: Drones in this category, such as the Harop and Skystriker, employ advanced guidance systems. (Please see the chart).

Footnote

[39] This combination results in exceptional precision and adaptability, enabling these drones to target effectively and engage objectives across various environments.

Footnote

[40] However, their high level of sophistication introduces complex challenges. The advanced technology and high cost of these drones raise concerns about unintended consequences.

Footnote

[41] While their precision is beneficial for military operations, it also increases the risk of civilian harm if the targeting algorithms fail, or if the drones are deployed in densely populated areas.

Footnote

[42] The opacity of the AI algorithms used in these drones further complicates accountability, as their decision-making processes may not always be transparent, potentially leading to breaches of IHL.

Footnote

[43]

Moderate Sophistication: Drones like the Switchblade 300 and Harpy fall into this category. They typically rely on a combination of guidance systems.

Footnote

[44] (Please see the chart) While these drones offer effective targeting capabilities, they may lack some of the advanced features and versatility found in more sophisticated systems.

Footnote

[45] As a result, their limitations can lead to less precise targeting, particularly in complex or unpredictable environments. The moderate sophistication of these drones makes them a cost-effective solution for many military applications. However, their operational limitations could reduce their effectiveness in challenging scenarios, potentially leading to higher risks of collateral damage, which could have severe consequences for civilian populations in conflict zones.

Footnote

[46]

Basic Sophistication: This category includes drones like the Shahed-136, which rely on simpler guiding systems such as basic GPS and GNSS.

Footnote

[47] These drones are generally less precise and offer limited flexibility, increasing the risk of collateral damage.

Footnote

[48] Despite their affordability and potential for mass deployment, their technological simplicity poses significant challenges, particularly in urban or densely populated areas. The lack of advanced targeting capabilities and guidance systems can result in unintended civilian harm, underscoring the need for careful strategic planning and consideration of their impact on civilian populations.

Footnote

[49] As a result, these drones require greater scrutiny, regulation, monitoring, and enforcement.

The increasing use of kamikaze drones by countries like Iran and Russia, which have developed less sophisticated yet widely deployable models, heightens the risks to civilians and civilian infrastructure.

Footnote

[50] This trend highlights the urgent need for robust international oversight and regulation to prevent misuse and ensure compliance with IHL. Whilst missiles and other remote warfare systems have long been subject to international regulation, the rapid proliferation of kamikaze drones introduces new challenges that existing frameworks are ill-equipped to address. The lack of effective regulatory responses to the rapid pace advancements in drone technology, particularly in terms of production and use, underscores the pressing need for comprehensive international regulations.

While higher-sophistication drone technology offers greater precision, it also introduces challenges related to transparency and potential for civilian casualties, if the technology fails, or is misused. In contrast, less sophisticated kamikaze drones, such as the Shahed-136, pose significant risks due to their inherent lack of precision, often resulting in greater collateral damage and casualties. These concerns underscore the urgent need for stringent regulation, monitoring, and oversight in the manufacture and use of all drones – regardless of their level of sophistication – to ensure they are produced and deployed responsibly and in full adherence to IHL.

Overall, while sophisticated AI-driven drones offer advanced capabilities and precision, they present significant legal and ethical challenges, particularly concerning accountability and compliance with IHL. Conversely, less sophisticated kamikaze drones, though cost-effective and easily accessible, present considerable risks due to their limited precision and higher potential for collateral harm. Addressing these varied risks is crucial for mitigating these issues and upholding humanitarian standards in modern warfare.

This theoretical overview sets the stage for a closer examination of a specific incident between Israel and Iran, used as an empirical case study to explore these issues in a concrete context. The study focuses on the use of the Shahed-136 drone in the attack on 13–14 April 2024. It analyses the technical aspects of the Shahed-136, assessing its sophistication, quality, and associated risks from both an engineering and legal perspective. This dual approach underscores the urgent need for a comprehensive international regulatory framework to address the evolving threats posed by drone technology and ensure compliance with international legal standards.

3. The problems with using basic technology in modern warfare: a case study of the Iranian Shahed-136 kamikaze drone

To grasp fully the broader implications and legal challenges posed by kamikaze drones in modern warfare – particularly those with less sophisticated systems – it is essential to address the unique risks they present. Analysing specific examples helps illustrate these challenges. This section critically examines the paradox of such drones: despite their seemingly rudimentary technological design, they often pose greater risks than more sophisticated drones, including imprecise targeting, indiscriminate strikes, and increased collateral damage.

Footnote

51

This study uses the Iranian-made Shahed-136 kamikaze drone as a key case study, given its extensive deployment since the onset of the Ukrainian-Russian war in 2022, as well as its use in the 2024 Iranian attack on Israel. Known for its limited sophistication, the Shahed-136 is primarily employed in kamikaze operations.

Footnote

52

By focusing on this drone, this study delves into the complexities posed by these seemingly less sophisticated weapons in modern combat scenarios. Understanding the technical workings of systems like the Shahed-136 provides legal scholars with crucial insights into their operation, allowing for a more informed assessment of the legal challenges, particularly the difficulties in ensuring compliance with IHL. Conversely, examining these drones from a legal perspective provides those involved in their design and operation, including engineers, with a framework to consider the broader ethical implications and legal standards that must be upheld in the development and deployment of such systems. This study is also intended to benefit policymakers and legislatures working on related matters. The interdisciplinary approach underscores the delicate balance between technological advancements and the ethical and legal

challenges of warfare, fostering a comprehensive understanding that benefits all stakeholders invested in these issues.

To illustrate these challenges, the Iranian-manufactured Shahed-136 is used as a case study to explore the issues involved. The Shahed-136 stands out as a significant subject for investigation due to its unique yet simple design, which exemplifies Iran's successful reverse engineering efforts.

Footnote

[53] Before its extensive use in the Russia-Ukraine conflict, detailed technical information about the Shahed-136 was limited. However, its deployment by Russia has revealed its reliance on basic to moderately advanced technology.

Footnote

[54] Developed by the Iran Aircraft Manufacturing Industrial Company (HESA), the Shahed-136 weighs a mere 200 kg, measures 3.5 m in length, and has a wingspan of 2.5 m. It boasts an operational endurance of up to 11.5 h and can operate at altitudes ranging from 60 to 4,000 m, with a flying range of 1,000 to 2,500 km, according to various sources.

Footnote

[55] Despite these capabilities, the Shahed-136 is relatively rudimentary compared to more sophisticated models like the IAI Harop.

Footnote

[56]

It should be noted that these drones are often not used individually; instead, they are frequently deployed in mass attacks, where hundreds of them are launched simultaneously in an attempt to overwhelm the defences of the target country.

Footnote

[57] Consequently, they are often used together, with the understanding that some will be sacrificed to allow others to breach the country's defences.

Footnote

[58] Their low cost and ease of acquisition facilitate this approach. As a result, there is a greater likelihood of unintended targets and civilians being hit. This practice constitutes a violation of IHL, as it does not seek specific military targets, but instead targets any available objective.

Footnote

[59]

When examining the technical characteristics of the Shahed-136, it is important to note that the drone primarily relies on conventional guidance technologies, specifically GNSS and INS.

Footnote

[60] Unlike more sophisticated drones, it does not utilise sophisticated algorithms driven by machine learning or artificial intelligence.

Footnote

[61] Such advanced algorithms can enhance a drone's effectiveness by enabling real-time adjustments to its flight path, improving target recognition, and optimising navigation in various environmental conditions.

Footnote

[62] In contrast, the Shahed-136's reliance on conventional guidance systems means it lacks these dynamic capabilities, which limits its precision and adaptability.

Footnote

[63] Consequently, it is less able to select targets, adjust its flight path, and ensure accuracy while in flight after being launched.

These limitations in the Shahed-136's guidance systems highlight the challenges it poses to precision and adherence to IHL, specifically the principles of distinction and proportionality.

Footnote

[64] Its reliance on conventional technologies like GNSS and INS complicates the differentiation between military and civilian targets, significantly increasing the risk of unintended collateral death, injury, and damage.

Footnote

[65]

The extensive use of cheap drones in the Ukraine-Russia conflict is another example that indicates how drones have radically changed the modern battlefield. More than 100 different varieties of drones have been used in this conflict. Old fashioned kamikaze drones like the Shahed-136, which require little technology, are used against civilian infrastructure, causing much damage and harm. First Person View (FPV) racing drones have been modified into remote-controlled kamikaze drones that cause huge numbers of deaths. For example, the UN Human Rights Monitoring Mission in Ukraine (HRMMU) reported that short drone strike campaigns in January 2025, caused many hundreds more people being killed or wounded than occurred by any other weapon.

Footnote

[66] Another example of drone use, is the case of the Houthis in Yemen who cannot afford sophisticated drones. They therefore deploy low-quality kamikaze drones that are indiscriminate in their targeting. The Qasef-1 kamikaze drone with its low fragmentation warhead control uses a crude and very unreliable GPS. These drones have been used indiscriminately in Yemen, Saudi Arabia and the UAE.

Footnote

[67] These examples indicate that although low-tech drones are cheap and simple to manufacture and use, they pose a global threat.

The design of these drones, which prioritises simplicity and cost-effectiveness over precision, often leads to broader targeting impacts and higher risks of collateral damage. In densely populated areas, this can result in substantial unintended civilian harm that outweighs the military benefits, making it difficult to maintain proportionality.

Footnote

[68] These challenges highlight the complexities of applying IHL in modern asymmetric warfare, where technologically limited weapons like the Shahed-136 complicate compliance with legal principles designed to protect civilians, mitigate unnecessary suffering, and ensure that military infrastructure is targeted. This underscores the need for enhanced legal frameworks, as well as manufacturing and operational guidelines, to address the evolving nature of warfare and ensure adherence to humanitarian standards.

The urgent need for improved legal and regulatory frameworks became evident during the intense conflict between Iran and Israel in April 2024, when Iran launched a large-scale offensive that included 170 drones, over 30 cruise missiles, and more than 120 ballistic missiles. These were launched not only from Iranian territory but also from Iraq, Syria, and Yemen throughout the night. While Iranian officials claimed that the Shahed-136 drones were intended to target Israeli military and economic infrastructure to weaken Israel, the drones' primary role appeared to be complicating the Israeli defence systems' ability to cope with the overwhelming number of projectiles.

Footnote

[69] Thus, as noted earlier, the drones seemed to be sacrificed to divert defence capabilities away from the two different types of missiles that were used.

Despite the extensive deployment of various types of weapons, most of the drones were intercepted before reaching Israeli territory.

Footnote

[70] Had the Shahed-136 drones been able to breach the Israeli defences, their relatively rudimentary design and concomitant inaccuracy (with an error margin of about 50 to 100 m from their intended targets) could have caused significant damage.

Footnote

[71] The imprecision of the technology, combined with the large-scale nature of the attack (170 drones), might have resulted in substantial collateral damage and significant civilian casualties had the drones managed to penetrate Israeli defences.

Footnote

[72]

In comparison, the new Shahed-238, an advanced iteration of the Shahed-136, exemplifies the evolving technological landscape. With a top speed of 500 to 520 km/h, the Shahed-238 represents a significant technological leap.

Footnote

[73] Deployed in attacks against Ukraine in 2024, the use of the Shahed-238 by Russia reflects the limitations of current international sanctions in curbing not only Iran's progress with this technology, but also the inability to restrain the activities of other countries that breach international law and misuse such advanced technologies.

Footnote

[74]

It is important to note that the downing of all 170 drones was an exceptional feat, one that would not typically be possible. In Israel's case, they were able to counter this complex threat by deploying a multi-layered defence strategy that included various systems, such as the Arrow, Iron Dome, and David's Sling. The Arrow system focused on intercepting ballistic missiles, while the Iron Dome and David's Sling targeted other short- and medium-range threats.

Footnote

[75] In addition to their own defence systems, Israel received support from the U.S., Britain, France, and Jordan, which deployed anti-aircraft batteries, fighter jets, as well as guided and cruise missiles to engage accurately and destroy the incoming drones.

Footnote

[76] While these systems were highly effective in neutralising the Shahed-136 drones, the operation required the involvement of multiple countries and came at a significant cost to intercept all the drones.

Footnote

[77]

This highlights a notable disparity between the relatively low cost of deploying the Shahed-136 drones and the much higher cost of intercepting them.

Footnote

[78] With a production cost of approximately $20,000 per unit, the Shahed-136 is a cost-effective option for asymmetric warfare.

Footnote

[79] This affordability, combined with its operational efficiency, significantly heightens its threat to both civilian and military targets.

Footnote

[80] In contrast, on the defence side, the high cost of intercepting these drones – around $3 million per missile – places a substantial financial burden on the defence systems used for interception.

Footnote

[81] This financial strain is exacerbated during swarm or blitz attacks, where multiple interceptors may be needed to neutralise a large number of drones deployed simultaneously.

Footnote

[82] This will have a significant impact on countries with limited resources. The economic imbalance underscores the strategic advantage for attackers using large numbers of low-cost drones simultaneously, highlighting the substantial financial burden placed on defending nations in such scenarios.

Footnote

[83] The high costs of interception not only strain defence budgets, but also raise various humanitarian concerns, as they intensify the military challenges faced by nations with constrained resources.

Footnote

[84]

Regarding efforts to restrain countries like Iran and North Korea, the ongoing evolution of technology used in the Shahed-136 drone reveals the shortcomings of the current sanctions regime.

Footnote

[85] This highlights the urgent need for more robust measures to monitor and control the proliferation of such technologies. For example, the Shahed- 136's MD550 engine, modelled after the German L550, and manufactured entirely within Iran, demonstrates the country's capability to produce sophisticated military technology despite stringent international sanctions.

Footnote

[86] Whilst most components of the Shahed-136 are imported, with the exception of the MD550 engine and the Nasir anti-jammer circuit, Iran's ability to source

critical parts from both Western and East Asian countries exposes a significant flaw in the global sanctions framework.

Footnote

[87] The gaps in the enforcement of the sanctions regime not only enable Iran to continue producing these drones, but also allow it to supply them to its allies and proxies, thereby extending its influence and destabilising various conflict-prone regions.

Footnote

[88]

Given these insights, it is crucial to recognise that while potential technological advancements could improve the Shahed-136's precision and operational capabilities, its current design prioritises simplicity and cost-effectiveness to boost sales.

Footnote

[89] This focus on affordability underscores the urgent need for a comprehensive approach to regulating and monitoring emerging technologies, such as drones, to ensure that advancements in weapon production and use comply with international standards and humanitarian principles.

To address these challenges and the ongoing evolution of drone technology, a critical examination of potential technological enhancements for the Shahed-136 highlights several key areas for improvement:

- **Advanced GNSS Techniques:** Integrating Differential GNSS (DGNSS) and Real-Time Kinematic (RTK) GNSS could significantly enhance the Shahed-136's targeting accuracy.
- Footnote
- [90] DGNSS provides corrections to GNSS signals, reducing positional errors to within a meter or even a few centimetres,
- Footnote

- [91] while RTK GNSS offers real-time updates that further improve precision. By employing these advanced GNSS techniques, the accuracy issues currently caused by signal obstruction and interference could be mitigated.
- Footnote
- [92]
- **Improved INS:** Enhancing the INS with high-precision sensors and advanced sensor fusion techniques could mitigate the limitations of sensor drift and the absence of GNSS updates.
- Footnote
- [93] An improved INS would provide more stable and accurate navigation data, even in the absence of GNSS signals, thereby increasing the drone's overall precision.
- **Optical and Radar Targeting Systems:** Incorporating Electro-Optical/Infrared (EO/IR) sensors and radar systems would greatly enhance the Shahed-136's target recognition and tracking capabilities.
- Footnote
- [94] EO/IR sensors would improve target identification in various conditions, while radar systems would boost detection and targeting accuracy, particularly in challenging environments.
- Footnote
- [95]
- **AI-Driven Guidance Systems:** Implementing AI-driven guidance systems would enable real-time adjustments to the drone's trajectory based on environmental changes and target movement.
- Footnote
- [96] AI would facilitate more precise targeting and reduce the likelihood of collateral damage by allowing dynamic corrections during flight.
- Footnote
- [97]

These advancements could be easily incorporated into the Shahed-136 and similar kamikaze drones, significantly enhancing their operational capabilities. They would address the current limitations and help mitigate the risk of

indiscriminate attacks. However, the focus on simplicity and cost-effectiveness in its design makes the incorporation of such advanced features unlikely, highlighting the complexities of balancing technological progress with affordability in military applications.

The following section provides a thorough legal analysis of the Shahed-136, focusing on its compliance with IHL principles. This examination considers the broader implications for international law and proposes necessary reforms to address the challenges posed by such technologically basic, yet strategically significant, weapons. By integrating technical and legal perspectives, this study offers a comprehensive understanding of how emerging warfare technologies can be effectively regulated while upholding humanitarian standards.

4. The cost and complexity dilemma: how low-cost and basic drones challenge legal and ethical norms in warfare

The use of drones in warfare is not inherently illegal, but the rise of inexpensive models with less advanced systems, like the Shahed-136, is reshaping modern conflicts and presenting significant legal challenges. As noted in the previous sections, these drones often lack sophisticated targeting and guidance systems, causing them to land – or even crash–unpredictably. This unpredictability potentially violates the international law principles of distinction, proportionality, and necessity.

Footnote

98

This presents a significant issue, as low-cost drones with rudimentary systems are accessible to both state and non-state actors, raising urgent legal and ethical concerns. These concerns include the heightened risk of indiscriminate attacks and the erosion of accountability due to the difficulty in tracking who is using them, where, and how.

Footnote

[99] The Ukraine-Russia conflict, along with incidents like Iran's attack on Israel in April 2024, demonstrate how these less advanced technologies strain existing legal frameworks and contribute to unnecessary civilian casualties.

Footnote

[100]

Mass drone attacks and swarm tactics present specific threats, such as overwhelming air defences and forcing states to expend costly missile systems.

Footnote

[101] This creates a cost imbalance, where inexpensive drones are countered by expensive defences, fundamentally shifting the economics of modern warfare.

Footnote

[102] As drones become more prevalent, the distinction between acceptable and unacceptable combat methods blurs, influencing public perception and complicating future legal and ethical interventions. The normalisation of drone warfare will make it increasingly difficult to reverse these tactics.

States affected by drone attacks are calling for regulatory responses. However, there is scepticism about whether countries that have heavily invested in drone technology will support greater control and stricter regulations, as these nations often prioritise offensive capabilities over human life. Urgent international debate and consideration are necessary to address the challenges posed by drones with less advanced systems. Without effective legal frameworks and accountability mechanisms, unchecked drone technology could fundamentally alter conflict dynamics, eroding ethical boundaries and undermining international, and even domestic, legal standards.

The following section will examine the legal implications of deploying less advanced drones and the broader challenges they pose to established legal frameworks.

4.1. The challenge to international humanitarian law

IHL, particularly the Geneva Conventions, is designed to mitigate the impact of armed conflict on societies, with a focus on protecting non-combatants, such as civilians.

Footnote

[103] Central to IHL are the principles of distinction, proportionality, and necessity – principles that are increasingly undermined by the use of less sophisticated, inexpensive drones in conflict zones.

Firstly, the principle of distinction requires combatants to differentiate between military targets and civilian entities to protect non-combatants from harm.

Footnote

[104] Due to their limited precision and targeting capabilities, these basic drones often fail to uphold this principle, either entirely or effectively at times, especially when compared to more sophisticated military drones.

Footnote

[105] As mentioned earlier, these drones generally lack the advanced recognition systems needed to identify accurately and distinguish between legitimate military targets and non-combatants, including civilians. As a result, their use in situations where proper identification is not made can lead to indiscriminate attacks and increased collateral damage. Evidence from various investigations reveals a troubling trend: rather than reducing civilian casualties as often claimed, low-cost drones have been linked to an increase in such casualties.

Footnote

[106] Moreover, a 2020 report indicates that governments frequently underreport civilian deaths, misidentifying civilians as combatants and failing to fully account for collateral damage.

Footnote

[107] Comparative studies further underscore the gravity of this issues. For instance, it is reported that traditional airstrikes, where people are in the cockpit, result in one civilian death per 21 airstrikes, whereas drone strikes lead to one civilian death per 1.6 attacks.

Footnote

[108] This stark disparity highlights the heightened risk of civilian harm associated with low-cost drones.

This is deeply concerning, as the principle of necessity mandates that military actions be directed toward legitimate military objectives, with civilian harm minimised and kept proportionate.

Footnote

[109] Basic drones, with their limited sophistication, undermine this principle by often failing to meet the precision and operational standards required to ensure that their use is both necessary and proportionate.

Footnote

[110] The indiscriminate nature of many low-cost drones appears to result in excessive civilian casualties, at times outweighing the intended military advantage, thereby violating the principle of necessity.

Moreover, the principle of proportionality requires that any attack carefully balance the anticipated military gain against the potential harm to non-military targets and non-combatants. As noted earlier, due to their inherent lack of precision, low-cost drones often struggle to comply with this principle. Their inability to target accurately military objectives whilst avoiding civilian harm frequently results in disproportionate damage to people and places that should be protected from warfare.

Footnote

[111] This risk is further heightened by the fact that these drones are often deployed in situations where precision is essential, yet their limited capabilities

make it difficult to ensure that the harm they cause remains proportionate to the military benefit.

Footnote

[112]

Thus, the challenges posed by less sophisticated drones to the IHL principles of distinction, necessity, and proportionality highlight significant gaps in the current legal framework.

Footnote

[113] While those using such drones may argue that they take necessary steps to avoid harm to non-military targets and non-combatants, the reality is that these steps are often ineffective due to the inherent limitations of these weapons.

Footnote

[114] Consequently, these drones, with their limited precision and targeting capabilities, contribute to increased damage to civilian infrastructure and causalities. They fail to meet the operational standards required for lawful military conduct. However, the issues surrounding these drones extend beyond their effectiveness in conflict.

The deployment of both low-cost and sophisticated drones raises critical issues concerning attribution, responsibility, and accountability. The challenges in tracing and identifying the perpetrators of drone strikes, whether involving rudimentary, or advanced systems, expose significant shortcomings in how international law addresses modern warfare. Low-cost drones, due to their ease of use, obscure the attribution of responsibility, while sophisticated drones equipped with AI introduce a host of additional, complex accountability challenges.

Addressing these issues requires a comprehensive review and updating of the international legal framework to align it better with the realities of contemporary drone warfare. The Geneva Conventions must either address these challenges directly, or a new protocol is needed. Establishing clear guidelines to ensure accountability and responsibility is crucial for upholding international justice and humanitarian standards. As drone technology and usage evolve rapidly, so too

must the legal structures and rules, to manage effectively the profound implications for warfare, humanitarian law, and human rights protection.

4.2. Attribution of responsibility and accountability

Another major legal challenge arising from the use of both low-cost and sophisticated drones in warfare is the issue of attribution, responsibility, and accountability. Determining who is legally responsible for drone actions and ensuring they are held accountable is crucial. These challenges expose significant gaps in the current legal frameworks regulating drone use in conflicts.

Low-cost drones, in particular, due to their limited technology and affordability, make it difficult to trace and assign responsibility for the damage they cause. These drones are often small, lightweight, and difficult to trace, enabling unidentifiable attacks. Unlike traditional warfare, where state actors and the weapons they use are typically identifiable, the use of inexpensive drones allows individuals, non-state actors, and militias to operate with relative anonymity.

Footnote

[115] This obscurity complicates the identification of those responsible for violations.

The increasing accessibility of drones not only raises accountability issues for states, but also empowers non-state actors and insurgent groups, who often disregard international norms.

Footnote

[116] This shift underscores the urgent need for updated legal frameworks to address the complexities posed by both low-cost and sophisticated drones.

These attribution challenges complicate the application of international law, making it difficult to hold perpetrators accountable for war crimes or other breaches of international norms. The elimination of individuals through drone strikes, while potentially effective, often bypasses due process and reduces complex geopolitical and ethical questions to mere acts of removal.

In contrast, high-tech drones equipped with advanced algorithms and AI for target identification present their own set of accountability issues. These sophisticated drones operate autonomously, making decisions based on complex AI systems. This diffuses responsibility across multiple actors, including manufacturers, operators, and AI programmers, creating accountability gaps that are difficult to address. Reliance on AI-driven decision-making can result in unintended consequences or errors, complicating efforts to ensure accountability. Traditional mechanisms for assigning responsibility and delivering justice are therefore ill-equipped to address the complexities of AI-driven warfare. These challenges underscore the need to update the applicable legal framework.

Effective attribution and accountability are essential for upholding the legitimacy of international justice and enforcing humanitarian laws. As drone technology advances, legal structures must evolve to ensure that violations of IHL, in particular, can be addressed, with clear guidelines and mechanisms to navigate the changing landscape of drone warfare.

4.3 Empowering non-state actors and terrorist groups

A critical issue to consider when assessing the need to update international law is the accessibility and affordability of drones, which empowers not only states but also non-state actors, insurgent groups, and even terrorist organisations.

Footnote

[117] This further complicates the legal landscape of modern warfare. Groups like Hezbollah have already demonstrated the potential of using inexpensive, basic drones to conduct attacks, surveil enemy movements, and spread propaganda.

Footnote

[118] The ethical and legal implications of providing such powerful tools to actors with malicious intent are profound and must be carefully assessed.

This is particularly important because when non-state actors use drones, they undermine traditional ethical boundaries of warfare, where states are the primary actors and are expected to follow established norms, such as protecting civilians and adhering to the laws of war.

Footnote

[119] Non-state actors, in contrast, are not bound by the same ethical constraints and often violate international laws that prohibit the intentional targeting of civilians.

Footnote

[120] The use of drones by these groups exacerbates these issues, further blurring the distinction between combatants and civilians, as well as between legitimate and illegitimate actors and actions in warfare.

Footnote

[121]

These challenges are evident in Iran's proliferation of drones to its proxy militias across the Middle East. In recent years, Tehran has supplied these proxies with a substantial number of less sophisticated drones and has also facilitated their local production.

Footnote

[122] This capability has not only enhanced the operational effectiveness of these militias, but also enabled them to inflict significant damage, drastically altering the dynamics of conflicts in the region. While these drones may be less sophisticated compared to other military systems, they are still capable of posing substantial risks and causing considerable harm to both military targets and civilian infrastructure.

The widespread use of these drones by non-state actors highlights a troubling trend of increased civilian casualties and collateral damage. As these groups exploit the technology's low cost and ease of use, the legal and humanitarian concerns become even more urgent.

4.4 Increased civilian casualties and collateral damage

As previously noted, a particularly troubling legal aspect of drone warfare is the potential for increased civilian casualties and collateral damage. The lack of precision in cheap, basic drones significantly raises the risk of hitting unintended targets, especially in densely populated areas where the danger to civilians is amplified.

Footnote

[123] The legal obligation to minimise harm to non-combatants is often compromised when these types of drones are deployed in such environments. This has been tragically demonstrated in the ongoing war in Ukraine, where Russia has used Iranian-made kamikaze drones, particularly the Shahed series, to target critical civilian infrastructure–including water supplies, hospitals, and electricity–placing tens of thousands of civilian lives at risk.

The violation of civilians' rights in these conflicts extends beyond death and the physical destruction of infrastructure.

Footnote

[124] As previously noted, the deficiencies in the guidance systems and technological limitations of these drones prevent them from reliably distinguishing between combatants and civilians. Their low cost makes them easy to deploy in large numbers, increasing the potential for misuse by both state and non-state actors. Consequently, not only have many lives been lost, but essential infrastructure has also been severely damaged, resulting in long-term suffering for entire communities.

Footnote

[125]

Part of the issues lies in the ability to conduct operations covertly and without attribution, which enables operators to act more recklessly and carry out indiscriminate attacks with less fear of identification, retaliation, or accountability. This, coupled with the low cost of the technology, encourages more frequent use and risk-taking. Operators are more likely to deploy drones in situations where the potential for collateral damage is high. Such actions undermine the principle

of proportionality, which dictates that the harm caused by an attack should not exceed the military advantage gained.

Footnote

[126] In the case of low-cost drones, these legal considerations are often disregarded, resulting in unnecessary and widespread damage.

The harm caused by these drones extends beyond physical damage and death. It also severely impacts the psychological and mental well-being of civilians living in conflict zones. The psychological effects, particularly on children, are well-documented.

Footnote

[127] Survivors of drone strikes often describe the buzzing sound of an approaching drone as their only warning, leaving them little time to escape.

Footnote

[128] Unlike traditional warfare, where there may be some forewarning as fighting draws near, drone strikes offer no such opportunity.

Footnote

[129] This unpredictability makes drone warfare especially devastating for civilians, who live in constant fear of an attack they cannot anticipate or avoid.

The constant presence of drones, even without direct strikes, instils a pervasive and ongoing sense of fear. This disruption not only affects their daily lives, but also infringes on children's right to education.

Footnote

[130] Prolonged exposure to drones has been linked to severe mental health issues, including anxiety, depression, and insomnia.

Footnote

[131]

4.5 Normalisation of low-cost drone warfare and public perception

As mentioned, a growing concern with the rise of basic drones in warfare is their potential to alter the traditional boundaries and ethical and legal standards of combat. A major problem is that the use of drones may become normalised, setting a new standard for remote and often irresponsible warfare.

Footnote

[132] The more affordable and accessible these drones are to both state and non-state actors, the more the general public may become desensitised to them as a part of modern warfare.

Footnote

[133] In the past decade, the widespread use of inexpensive, low-tech drones in conflict zones like Yemen, Afghanistan, and Pakistan has already shaped public perceptions, making drone warfare increasingly politically acceptable.

Footnote

[134] This trend is particularly evident in conflicts such as those between Iran and Israel, as well as Ukraine and Russia, where drones have become a central component of conventional warfare. If the use of drones continues to spread without proper legal regulation, the traditional principles of acceptable warfare may gradually erode. Over time, the public may become indifferent to the proportionality of actions taken during conflicts and whether they comply with international laws concerning drone use. This normalisation would make it difficult in the future to advocate for legal reform, or implement stricter regulations. This is because reversing the public perception that it has developed is challenging. Already, the general population's attitude towards the issue of drone usage is gradually shifting toward apathy.

Footnote

[135] This results in less incentive to restrict or even outlaw these tools. This could lead to a future where military operations are increasingly unbound by notions of accountability, accuracy, or ethics, as the tools of war become cheaper, more accessible, and less likely to be constrained.

5. Conclusion

This article has explored the growing usage of inexpensive, unsophisticated drones around the world and their profound impact on conflict zones, particularly on civilian lives. These drones have been deployed in areas such as Russia and Ukraine, as well as Iran and Israel, and have spread across the Middle East and other regions. They have become easily accessible to both state and non-state actors, who exploit their affordability and availability, often without hesitation. It is particularly low-cost, less sophisticated drones that have become major tools for various actors, as the study indicates. The widespread use of these unsophisticated drones significantly increases the risk of indiscriminate attacks, violating the principle of distinction and other IHL doctrines intended to protect non-combatants in conflict zones.

The detailed analysis in this article reveals that the risks posed by these systems extend beyond physical harm, profoundly affecting the psychological well-being of civilians. As demonstrated, the cheaper and more rudimentary the drone, the more likely it is to amplify risks to civilian populations. While technical improvements to enhance their precision are theoretically possible, manufacturers intentionally limit these developments to maintain the low cost and accessibility that make these drones attractive for misuse.

Initially seen as tools of remote warfare with the potential to protect civilians through precision, recent trends show a troubling shift. In the case of Iran and Israel, for example, drones are frequently deployed in large numbers, or as swarms, often without specific, or legitimate targets. Their primary objective is to overwhelm enemy defence systems rather than to strike precise military targets. This significantly increases the risk of indiscriminate attacks, resulting in civilian casualties and property damage, causing unnecessary suffering. Such tactics heightens the ethical concerns associated with the use of drones in modern warfare. The international community’s slow response to regulating drones is not merely an oversight, but a consequence of conflicting national interests. Some

states benefit from drone proliferation through exports and their own military application. The lack of a unified stance on drone control has resulted in weak enforcement mechanisms and the continued, unchecked expansion of drone use. The ongoing silence from the international community on this matter, with each new deployment of these systems, will normalise their use, creating a dangerous precedent of acceptance. The fact is that even the most sophisticated drones, including those with AI, are perceived as a significant threat to peace and security due to their opaque and autonomous nature. This undermines the principles of IHL and erodes accountability.

This paper concludes that, in the absence of solid international oversight and regulation, the unabated proliferation of those drones will continue to threaten civilian lives and undermine peace and security worldwide. Addressing this issue requires overcoming entrenched political and economic interests that have stalled past regulatory efforts. Without a shift in global priorities, existing gaps in drone governance will persist, further destabilising conflict-prone regions.

The consequences are even more profound: to the extent that these drones become further integrated into conflict strategies, they have the potential to destabilise regions already prone to violence and conflict, further exacerbating humanitarian crises and eroding principles of coexistence and security outlined in international conventions. Therefore, it is imperative that steps are taken in the short term before these weapons of war become irreversible staples of modern conflict. Upgrading the international legal framework to regulate drones is crucial. It is essential to securing international peace and security at a time when conflict and human rights and humanitarian law violations are on the rise globally.

Over the last few days, credible reports indicate that Iran managed to hit several American radar systems in the Gulf region.

The most prominent, and so far the only confirmed hit, struck the AN/TPY-2 radar at the Muwaffaq Salti Air Base in Jordan, which serves as the primary sensor for the THAAD missile defense system. Iran is also

reported to have damaged the AN/FPS-132 early-warning radar at the U.S. base in Al-Udeid, Qatar, and may have targeted additional sensor sites across the region.

While the exact circumstances remain unconfirmed, early reports suggest these attacks were carried out primarily with Shahed-type long-range drones. This raises the question of why the United States and its allies were so unprepared for this type of threat and how relatively simple systems can inflict hundreds of millions of dollars in damage across separate strikes.

This post discusses why these strikes were able to succeed and how states can posture in the future to prepare effectively for large-scale, long-range drone threats.

The long-range drone challenge in the Middle East

In principle, Iranian long-range drones, including the Shahed series, should be relatively easy to defeat. They are slow, follow relatively predictable routes, and lack advanced stealth features. Yet several

structural factors make long-range defense in the Gulf more difficult than it might first appear.

Before the war, regional missile defense was highly optimized for ballistic missile defense. The Gulf states invested billions in sophisticated early-warning, tracking, and engagement solutions to address what was seen as Iran's primary threat, namely ballistic missiles.

Existing infrastructure is, of course, not useless, but it is also not well-suited to the long-range drone threat. Shahed-type systems can approach their targets at relatively low altitudes, sometimes only tens of meters above the ground. Detecting these types of targets can be difficult for non-optimized radars because low-altitude flight places them within the radar horizon and against heavy ground clutter.

Many of the region's existing radars were instead optimized for long-range air surveillance, pointing up into the sky rather than for persistent low-altitude coverage. What results is not an absence of capable sensors, but coverage gaps and suboptimal detection geometry against slow and low-flying drones. Alternative sensor solutions, such as

the acoustics-based early-warning network deployed by Ukraine, do not exist in the region.

Pre-war hubris?

Part of the explanation for why Iranian long-range drones have been effective, paradoxically, probably also lies in Israel's success in intercepting Shahed-type drones during the 2024 and 2025 conflict episodes. Here, Israel was able to intercept hundreds of drones repeatedly, without a single drone reaching its target, let alone causing serious damage.

That success, however, was heavily shaped by geography. The long distance between Iran and Israel led to travel times of up to 10 hours, while Israel's small size created highly predictable attack vectors, simplifying interception. In fact, Israel and allied aircraft intercepted most incoming long-range drones long before they arrived in Israeli airspace, notably inside Jordan.

With the conflict now extending to Iran's immediate neighbors, these geographic advantages no longer apply. Shahed travel times are much shorter, while the Gulf's large geographic expanse creates many more potential attack vectors that must be defended simultaneously. As a result, while most long-range drones are still intercepted, some inevitably slip past defensive systems.

Existing long-range drone defenses

At present, the United States and the Gulf states appear to rely primarily on manned aircraft to intercept incoming long-range drones. This approach makes sense given the region's geographic realities, as only large numbers of patrolling or scrambled aircraft can provide sufficient levels of area defense.

This initial layer appears capable of intercepting the vast majority of incoming drones, sometimes at relatively low cost. For example, intercepting incoming Shaheds with rotary-wing aircraft, such as the

AH-64 Apache, costs roughly 5,000 to 10,000 USD per flight hour, plus a few thousand dollars for gun ammunition.

Even using an F-15E fighter jet, which costs up to 30,000 USD per flight hour, together with an AIM-9X Sidewinder priced at roughly 400,000 to 500,000 USD, is arguably acceptable given the economic advantage of the United States and its regional partners, as well as the potential cost of repairing or replacing whatever infrastructure might otherwise be struck.

That said, most fighter jets currently appear to use APKWS missiles to combat long-range drones, which cost only about $30,000 each. This greatly improves the cost-effectiveness of these missions and renders cost concerns largely moot.

While manned aircraft have proved effective, some long-range drones slip through this initial layer. If that happens, surface-to-air missile systems provide a second layer. As noted by many commentators, interception attempts at this layer can be very expensive, given that the Gulf states do not deploy suitable point-defense systems for low- or

lower-cost engagements, while the United States has only a limited number of suitable assets in the region, such as C-RAM.

Whether employing these more expensive systems is cost-inefficient is not always clear-cut. The United States would likely have been willing to allocate dozens of Patriot interceptors to prevent hits on its AN/TPY-2 and AN/FPS-132 radars. That said, regardless of whether the defended target justifies the cost, the engagement attempt will be expensive, and the expended munitions, given long production lead times, will be difficult to replace.

Pre-war reporting suggests that the Gulf states, together with the United States, were interested in acquiring dedicated counter-UAS systems that could have effectively complemented the upper missile defense layers to defend against long-range drones. However, procurement processes proved too lengthy, creating the now obvious capability gap at the start of the war.

It is also worth remembering that both Jordan and Qatar gave up their fleets of 60 and 15 Gepard systems in 2023, when Germany and the

United States purchased them for transfer to Ukraine. Both countries could have made good use of these systems in the ongoing war.

How to defend long-range drone threats

Despite these challenges, the ongoing Iran War, as well as the war in Ukraine, increasingly clarify what a future missile defense posture designed to address a large-scale long-range drone threat must look like, including in Europe.

First, states must be able to establish an effective first layer of airborne area defense. Manned aircraft are broadly suitable, though large fleets of unmanned aircraft, with lower maintenance and per-flight-hour costs, combined with lower-cost air-to-air missile capabilities, would be preferable. Several low-cost ground-to-air interceptors currently under development in Europe and beyond could, in principle, be adapted for this role.

Second, states need to deploy affordable point-defense systems with a low cost per interception to deal with long-range drones that slip through the initial layer. These systems will necessarily be point-defense

in nature, as costs tend to increase with engagement range. In other words, the objective is not to create a second area defense layer, which is not feasible, but rather to establish a redundancy layer for critical targets. How many targets can be defended in this way depends on the level of investment a state is willing to make.

Third, since it may not be feasible for a state to provide dedicated ground-based missile defense for every potential high-value target, the public sector may need support from the private sector. Just as discussions are underway about whether airports should be able to counter small quadcopter-type drones that threaten civilian air traffic, a broader debate may be required about whether large industrial actors and companies — under specific circumstances — should be authorized, or even required, to acquire defensive capabilities designed to counter larger and more potent drone threats.

The objective is not to privatize defense, but to recognize that as drone threats proliferate and civilian infrastructure becomes an increasingly common target for drone strikes in war, the military alone may not be

able to provide the level of protection required across all sectors. This may be particularly relevant for countries such as Norway, which are geographically large and host significant numbers of dispersed critical infrastructure sites. Needless to say, this would require major policy reforms.

Ultimately, it must also be acknowledged that a perfect drone defense solution capable of intercepting every incoming system at all times and in all places does not exist.

Ukraine's experience with drone defense shows that even with extensive operational experience, gaps can still emerge. Even on days when "only" 90 Russian long-range drones arrive, operator error, chance, technical problems, or other factors allow individual drones to get through. That is the nature of air defense. For this reason, defensive components must always be complemented by other capabilities, ideally to make clear to an adversary that an attack is not worthwhile and to retain the ability to generate effects within enemy territory should deterrence fail.

Drones and the Cost-Exchange Challenge in Modern Warfare

September 15, 2025 by Miriam McNabb Leave a Comment

When cheap drones force expensive defenses, the balance of power shifts

Modern warfare is in the midst of a profound change. Small, inexpensive drones are reshaping the battlefield, forcing militaries to rethink how they fight and, just as importantly, how much they spend. The problem of the negative cost-exchange ratio, when defending against a low-cost threat requires disproportionately expensive responses, is becoming one of the defining challenges of our time.

Recent events in Poland, lessons from Ukraine and Russia, and policy shifts in the United States illustrate how urgent this issue has become. The way the global community responds may determine the character of future conflicts.

Poland's Wake-Up Call: Expensive Missiles Against Cheap Drones

A recent incident in Poland demonstrates the imbalance clearly. Reports indicate that inexpensive Russian "Gerbera" drones, some built of simple materials like foam and plywood, entered Polish airspace in what appeared to be a decoy mission. In response, Poland and its NATO allies activated advanced defenses, deploying fighter jets, surveillance systems, and Patriot missile batteries. Some of the intercepting missiles cost millions of dollars, while the drones they destroyed were worth only a fraction of that amount.

This episode underscores the core issue. If every cheap aerial incursion requires a high-value missile in return, the defender risks being bled dry financially. Ukraine's President Volodymyr Zelenskiy has highlighted the unsustainable nature of such exchanges, urging partners to rethink air defense strategies in light of this dynamic.

Ukraine and Russia: A Shifting Balance

The conflict between Russia and Ukraine illustrates both sides of the cost-exchange equation. Early in the war, Ukraine used small drones to target and disable expensive Russian tanks and armored vehicles. Inexpensive systems were able to achieve significant battlefield effects against some of the most costly pieces of equipment in Russia's arsenal.

Russia has since adapted, learning from its early setbacks. Today, it emphasizes the rapid production of small, affordable drones that can be deployed in large numbers. These systems not only serve as weapons but also as decoys, saturating enemy defenses and forcing adversaries to expend high-value resources. In this way, Russia has inverted the problem, turning cost asymmetry into a weapon of strategy as well as tactics.

The United States Response

American policymakers and defense leaders have taken note of these lessons. Secretary of Defense Pete Hegseth issued a directive titled *Unleashing US Military Drone Dominance* that called for simplifying acquisition processes and accelerating deployment of drones across the force. The memo emphasized that adversaries are producing drones by the millions each year, while the United States has been slowed by red tape and outdated procurement practices.

The Department of Defense has also launched the Replicator Initiative, an effort designed to produce large numbers of attritable drones. These systems, intended to be affordable and scalable, can be used in greater volume without creating an unsustainable financial burden. Alongside Replicator, legislation and executive actions are focused on strengthening U.S. manufacturing capacity, expanding the industrial base for small uncrewed systems, and enabling smaller firms to scale production quickly.

Strategic Implications: Timing and Adaptation

The implications of these trends go far beyond individual incidents. Nations that continue to rely exclusively on expensive defensive systems for every engagement may find themselves at a disadvantage against adversaries that exploit the economics of cheap drones. Swarms of small systems, improvised designs, and decoys can all impose disproportionate costs, draining resources and creating strategic strain.

Western procurement processes, traditionally slow and risk-averse, face a moment of reckoning. The examples of Poland and Ukraine suggest that traditional responses, matching high-value threats with equally costly countermeasures, will not be enough in future conflicts. The pace at which countries adapt their acquisition systems, industrial capacity, and defense strategies may prove decisive.

This is not only a question for the United States or Europe. Other global powers are closely watching these developments, aware that the balance between cost and capability may shape the outcome of future wars. The credibility of deterrence, and the ability to maintain military effectiveness, may increasingly depend on how fast governments can adopt affordable, scalable, and flexible defensive systems.

The Shift is Here

The Poland incident, the evolution of tactics in Ukraine, and U.S. efforts to reform procurement and boost manufacturing all point to the same conclusion. Modern warfare is no longer only about technological superiority, it is about economic sustainability. Inexpensive drones are changing the terms of engagement, forcing defenders into unfavorable cost calculations.

The United States has begun to respond with new policies and initiatives, but the challenge extends globally. How quickly nations learn the lessons of cost asymmetry, and how effectively they adapt, will shape the wars of tomorrow. In an era where cost is as decisive as capability, the ability to restore balance in the cost-exchange ratio may determine which nations hold the advantage in the next conflict, wherever it arises.